

Approximation Algorithms

Unified Question–Answer Notes: Exercises 35.1 and 35.2

Core Facts

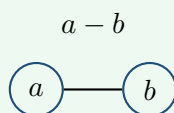
Vertex Cover	Pick an uncovered edge (u, v) , add both endpoints. Chosen edges form a matching.
Metric TSP	Triangle inequality: $c(u, w) \leq c(u, v) + c(v, w)$.
2-Approx VC	$ C = 2 A \leq 2\text{OPT}$.
2-Approx Metric TSP	$c(\text{tour}) \leq 2c(\text{MST}) \leq 2\text{OPT}$.

Question 35.1-1

Give an example of a graph for which APPROX-VERTEX-COVER always yields a suboptimal solution.

Answer

Take one edge:



The optimum chooses one endpoint:

$$\text{OPT} = 1.$$

APPROX-VERTEX-COVER chooses the only edge and adds both endpoints:

$$C = \{a, b\}, \quad |C| = 2.$$

Hence the algorithm is suboptimal. For k disjoint edges, it returns $2k$ while $\text{OPT} = k$, so the factor 2 is tight.

Question 35.1-2

Prove that the set of edges picked by APPROX-VERTEX-COVER forms a maximal matching.

Answer

Let A be the set of edges selected by the algorithm.

- 1. A is a matching.** After choosing (u, v) , every edge incident on u or v is removed. So no later chosen edge can share an endpoint with (u, v) .
- 2. A is maximal.** When the algorithm stops, every remaining edge has been removed. Any edge not in A shared an endpoint with some edge in A . Therefore no new edge can be added to A without breaking the matching property.

A is a maximal matching.

Remember: maximal = cannot add another edge; maximum = largest possible matching.

Question 35.1-3

The heuristic “repeatedly choose a highest-degree vertex” is proposed for Vertex Cover. Show that it need not have approximation ratio 2.

Answer

Use a bipartite graph $G = (R \cup L, E)$ with $|R| = n$ such that:

- R is a vertex cover, hence $\text{OPT} \leq n$;
- for each $i = 2, \dots, n$, create $\lfloor n/i \rfloor$ vertices in L , each of degree i ;
- connect them to currently lowest-degree vertices of R , so degrees in R stay nearly balanced;
- break ties in favor of L .

After all larger-degree vertices of L are selected, every remaining vertex of R has degree at most the current degree i , while some vertex of L still has degree i . Thus the heuristic can keep selecting vertices from L .

The number selected is at least

$$|L| \geq \sum_{i=2}^n \left\lfloor \frac{n}{i} \right\rfloor \geq n \sum_{i=2}^n \frac{1}{i} - (n-1) = \Omega(n \log n).$$

For sufficiently large n , $|L| > 2n \geq 2\text{OPT}$.

Highest-degree greedy is not a 2-approximation.

Question 35.1-4

Give an efficient greedy algorithm that finds an optimal vertex cover for a tree in linear time.

Answer

Algorithm

1. If the tree has no edge, return $\emptyset$.
2. Choose a leaf v and let u be its only neighbor.
3. Add u to the cover.
4. Delete u, v , and all edges incident on u .
5. Repeat on the remaining forest.

Why the choice is safe. Edge (u, v) must be covered. If an optimal cover contains v but not u , replace v by u :

$$C' = (C^* - \{v\}) \cup \{u\}.$$

Since v covers only (u, v) but u covers (u, v) and all other edges incident on u , C' is also optimal. So some optimal solution always contains u .

Apply the same argument repeatedly to the remaining forest.

With adjacency lists and a leaf queue, each vertex/edge is processed only a constant number of times:

$$O(|V| + |E|) = O(|V|) \text{ for a tree.}$$

Question 35.1-5

Minimum Vertex Cover in G is complementary to Maximum Clique in $\overline{G}$. Does a constant-factor approximation for Vertex Cover therefore give a constant-factor approximation for Clique?

Answer

No. Exact complements preserve optimal solutions, but not multiplicative approximation ratios.

If G has n vertices and minimum vertex-cover size k , then

$$\omega(\overline{G}) = n - k.$$

A 2-approximation for Vertex Cover may return $2k$, giving a clique of size

$$n - 2k.$$

But the optimum clique has size $n - k$.

Take

$$k = \frac{n}{2} - \varepsilon.$$

Then

$$\frac{\text{returned clique}}{\text{optimal clique}} = \frac{2\varepsilon}{n/2 + \varepsilon} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

So no positive constant approximation ratio follows.

Subtracting from n can destroy a multiplicative guarantee.

Question 35.2-1

In a complete undirected graph with at least three vertices, suppose the cost function satisfies the triangle inequality. Prove that every edge cost is nonnegative.

Answer

Assume some edge has negative cost:

$$c(u, v) < 0.$$

Choose a third vertex w .

Triangle inequality gives

$$c(u, w) \leq c(u, v) + c(v, w) < c(v, w).$$

Because the graph is undirected, $c(v, u) = c(u, v)$, so also

$$c(v, w) \leq c(v, u) + c(u, w) < c(u, w).$$

Thus

$$c(u, w) < c(v, w) \quad \text{and} \quad c(v, w) < c(u, w),$$

a contradiction.

$$c(u, v) \geq 0 \text{ for every edge.}$$

Question 35.2-2

Transform a TSP instance in polynomial time into one satisfying the triangle inequality while preserving the same optimal tours. Why does this not contradict the hardness of approximating general TSP?

Answer

Assume original costs are nonnegative. Choose

$$m > \max_{u \neq v} c(u, v)$$

and define

$$h(u, v) = c(u, v) + m.$$

Triangle inequality:

$$h(u, v) = c(u, v) + m < 2m,$$

while

$$h(u, w) + h(w, v) = c(u, w) + c(w, v) + 2m \geq 2m.$$

Hence $h(u, v) \leq h(u, w) + h(w, v)$.

Same optimum: every tour on n vertices uses exactly n edges, so every tour gets the same added cost nm :

$$H = C + nm.$$

Thus tour order is unchanged.

Why approximation does not transfer: if $H \leq \rho H^*$, then

$$C \leq \rho C^* + (\rho - 1)nm.$$

The extra additive term prevents a constant-factor guarantee for the original costs.

Question 35.2-3

Prove that the closest-point insertion heuristic for metric TSP is a 2-approximation.

Answer

When a new vertex u is chosen, it is the closest outside vertex to the current set. Connecting it to its closest current vertex v is exactly a Prim step. Therefore these chosen edges form an MST T . The insertion order gives a preorder traversal of T . For an optimal TSP tour H^* :

$$c(T) \leq c(H^*)$$

because deleting one edge from H^* leaves a spanning tree.

Walking every tree edge twice costs

$$2c(T).$$

Shortcut repeated vertices to get a Hamiltonian tour. Triangle inequality ensures shortcutting does not increase cost.

Therefore

$$c(H) \leq 2c(T) \leq 2c(H^*).$$

So the heuristic is a 2-approximation.

Question 35.2-4

For metric Bottleneck-TSP, give a polynomial-time approximation algorithm with ratio 3.

Answer

Algorithm

1. Compute a minimum bottleneck spanning tree T (an MST is sufficient).
2. Let B be the largest edge cost in T .
3. Order the vertices cyclically so consecutive vertices are at tree distance at most 3 (tree-cube property).
4. Join consecutive vertices to obtain a Hamiltonian cycle H .

Each edge of H shortcuts a tree path of at most 3 edges. By triangle inequality,

$$b(H) \leq 3B.$$

Let H^* be an optimal bottleneck tour with bottleneck value B^* . Deleting one edge of H^* gives a spanning tree whose maximum edge cost is at most B^* , so

$$B \leq B^*.$$

Hence

$$b(H) \leq 3B \leq 3B^*.$$

Thus this is a polynomial-time 3-approximation.

Question 35.2-5

For Euclidean TSP, prove that an optimal tour never crosses itself.

Answer

Suppose two tour edges (x_1, x_2) and (y_1, y_2) cross at an interior point P .

Replace them by the noncrossing pair (x_1, y_1) and (x_2, y_2) , reversing one path if necessary. The result is still a Hamiltonian tour.

Because P lies inside both crossing segments,

$$|x_1 y_1| < |x_1 P| + |P y_1|,$$

$$|x_2 y_2| < |x_2 P| + |P y_2|.$$

Adding,

$$|x_1 y_1| + |x_2 y_2| < |x_1 x_2| + |y_1 y_2|.$$

So uncrossing makes the tour strictly shorter, contradicting optimality.

An optimal Euclidean TSP tour has no crossing edges.